transistor, the emitter-base junction is forward biased by a DC power supply V_{EB} and the collector-base junction is reverse biased by the bias power supply V_{CB}. The forward bias decreases the depletion region across the emitter-base junction and the reverse bias increases the depletion region across the collector-base junction. Hence, the barrier potential across the emitter-base junction is decreased and that across the collector-base junction is increased. The voltage across the emitter-base junction is represented as V_{EB} and that across the collector-base junction as $V_{CB.}$

In an NPN transistor, the majority charge carriers in the emitter are electrons. As it is heavily doped, it has a large number of electrons. The forward bias across the emitter-base junction causes the electrons in the emitter region to flow towards the base region and constitutes the emitter current (I_E). The electrons after reaching the base region recombine with the holes in the base region. Since the base region is very narrow and lightly doped, the number of holes in it is not sufficient to recombine with electrons from emitter. Hence most of the electrons reach the collector region.

Eventually, the electrons that reach the collector region will be attracted by the collector terminal as it has positive potential and flow through the external circuit. This constitutes the collector current (I_C). The holes that are lost due to recombination in the base region are replaced by the positive potential of the bias voltage V_{BE} and constitute the base current (I_B). The magnitude of the base current will be in microamperes as against milliamperes for emitter and collector currents.

It is to be noted that if the emitter current is zero, then the collector current is almost zero. It is therefore imperative that a BJT is called a current controlled device. Applying Kirchoff's law, we can write the emitter current as the sum of the collector current and the base current.

$$I_E = I_B + I_C \qquad (10.1)$$

Since the base current is very small, we can write $I_E \approx I_C$. There is another component of collector current due to the thermally generated electrons called reverse saturation current, denoted as I_{CO}. This factor is temperature sensitive. Therefore, care must be taken towards the stability of the system at high temperatures.

The ratio of the collector current to the emitter current is called the forward current gain (α) of a transistor.

$$\alpha = \frac{I_C}{I_E} \qquad (10.2)$$

The α of a transistor is a measure of the quality of a transistor. Higher the value of α, better is the quality of the transistor. It means that the collector current is closer to the emitter current. The value of α is less than unity and it ranges from 0.95 to 0.99. This indicates that the collector current is 95% to 99% of the emitter current.

Working of a PNP transistor

The working of a PNP transistor is similar to that of the NPN transistor except for the

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